

JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-08

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TOPICS TO BE COVERED

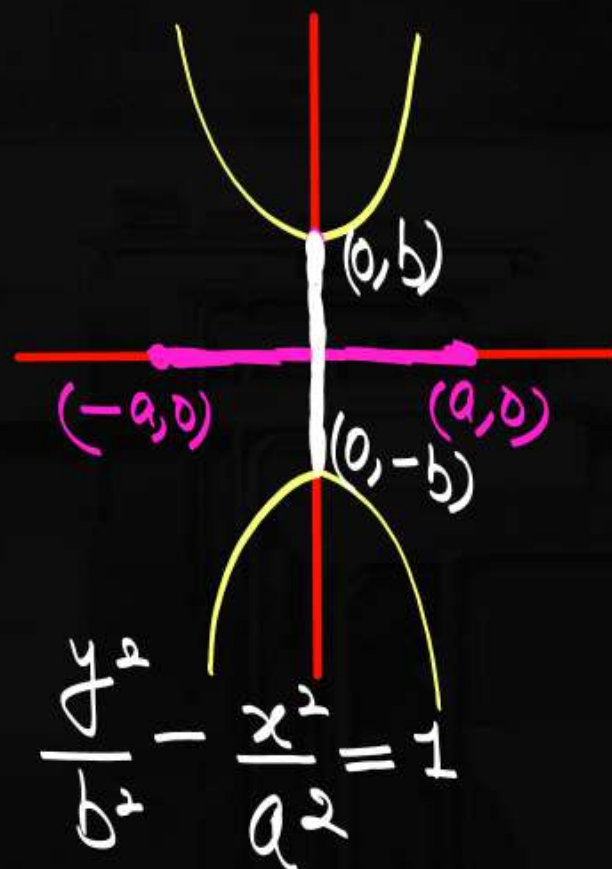
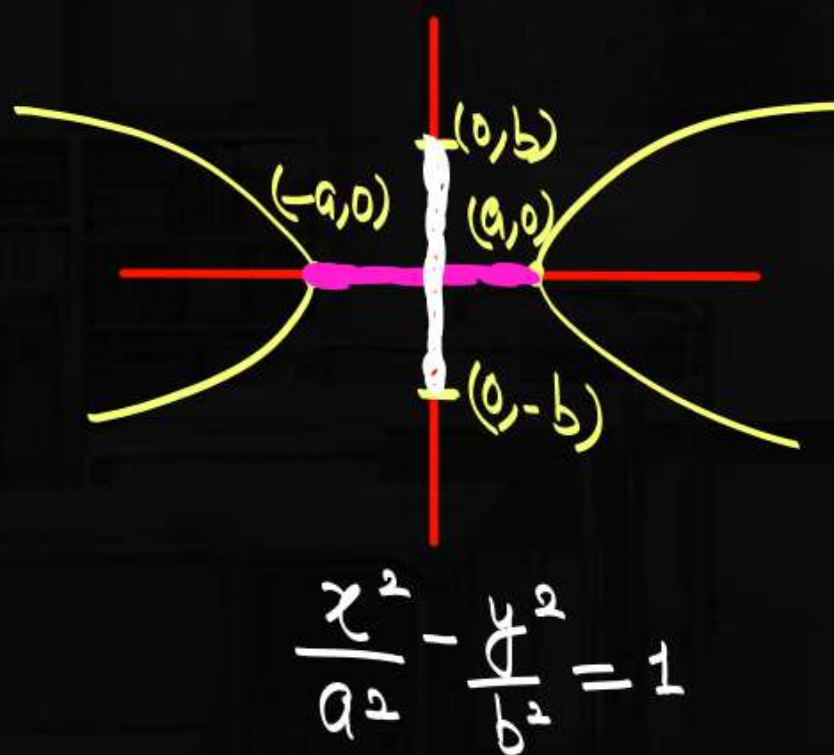
1

Exercise Discussion 10.4

2

Miscellaneous Exercise

★ **Conjugate Hyperbola:** The Hyperbola whose Transverse axis and Conjugate axis are respectively the Conjugate axis & Transverse axis of the given Hyperbola, is called the Conjugate Hyperbola of the given Hyperbola.



$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{--- (1)} \quad \& \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \quad \text{--- (2)}$$

are the Conjugate Hyperbolas.

Remark: Suppose that e & e' are respectively the eccentricities of (1) & (2). Then;

$$e = \frac{c}{a}, \quad e' = \frac{c'}{b}$$

$$\Rightarrow e^2 = \frac{a^2 + b^2}{a^2} \quad \& \quad e'^2 = \frac{b^2 + a^2}{b^2}$$

$$\Rightarrow \frac{1}{e^2} + \frac{1}{e'^2} = \frac{a^2}{a^2 + b^2} + \frac{b^2}{a^2 + b^2} = \frac{a^2 + b^2}{a^2 + b^2}$$

$$\Rightarrow \boxed{\frac{1}{e^2} + \frac{1}{e'^2} = 1}$$

Hence the result.

EXERCISE – 10.4

QUESTION

In each of the Exercises 1 to 6, find the **coordinates of the foci** and the **vertices**, the **eccentricity** and the **length of the latus rectum** of the hyperbolas.

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

(Sol) Given hyperbola is:

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$\Rightarrow \frac{x^2}{4^2} - \frac{y^2}{3^2} = 1$$

$$\text{So, } a = 4, b = 3$$

$$\Rightarrow c = \sqrt{4^2 + 3^2} = 5$$

$$\text{Therefore foci} = (\pm 5, 0)$$

$$\text{vertices} = (\pm 4, 0)$$

$$\text{eccentricity, } e = \frac{c}{a} = \frac{5}{4}$$

$$\begin{aligned} \text{length of latus Rectum} &= \frac{2b^2}{a} \\ &= \frac{2 \times 9}{4} = \frac{9}{2} \end{aligned}$$

QUESTION

In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$\frac{x^2}{9} - \frac{y^2}{27} = 1 \quad \text{p/w}$$

QUESTION

In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$9y^2 - 4x^2 = 36$$

HP/W

QUESTION

In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$16x^2 - 9y^2 = 576$$

hp/w

QUESTION

In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$5y^2 - 9x^2 = 36$$

HP/W

QUESTION

In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$49y^2 - 16x^2 = 784$$

hp/w

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Vertices $(\pm 2, 0)$, foci $(\pm 3, 0)$

(Sol) Here $a = 2$, $c = 3$

$$\Rightarrow c^2 = a^2 + b^2$$

$$\Rightarrow 9 = 4 + b^2$$

$$\Rightarrow b^2 = 5$$

Therefore the required hyperbola is:

$$\frac{x^2}{4} - \frac{y^2}{5} = 1 \Rightarrow 5x^2 - 4y^2 = 20$$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Vertices $(0, \pm 5)$, foci $(0, \pm 8)$

$$a = 5, \quad c = 8$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1.$$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

vertex .
 Foci $(0, \pm 3)$, foci $(0, \pm 5)$
 $a = 3, c = 5$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(0, \pm 13)$, the transverse axis is of length 8.

$$c = 13, \quad 2a = 8.$$

$$c = 13, \quad a = 4.$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1.$$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(0, \pm 13)$, the conjugate axis is of length 24.

(Sol) Given that $c = 13$ & $2b = 24$.

$$\Rightarrow c^2 = 169 \text{ & } b = 12.$$

$$\Rightarrow a^2 + b^2 = 169, \quad b = 12.$$

$$\Rightarrow a^2 + 144 = 169$$

$$a^2 = 25$$

$$\frac{y^2}{25} - \frac{x^2}{144} = 1.$$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(\pm 3\sqrt{5}, 0)$, the latus rectum is of length 8.

(Sol) Given that $c = 3\sqrt{5}$.
 $\Rightarrow c^2 = (3\sqrt{5})^2 = 9 \times 5$
 $\Rightarrow \boxed{a^2 + b^2 = 45} \text{ --- (1)}$

Also, length of latus Rectum = 8

$$\Rightarrow \frac{2b^2}{a} = 8$$

$$\Rightarrow \boxed{b^2 = 4a} \text{ --- (2)}$$

Using (2) in (1), we get;

$$a^2 + 4a = 45$$
$$\Rightarrow a^2 + 4a - 45 = 0$$

$$\Rightarrow a^2 + 9a - 5a - 45 = 0$$

$$\Rightarrow a(a+9) - 5(a+9) = 0$$

$$\Rightarrow (a+9)(a-5) = 0$$

$$\Rightarrow a - 5 = 0$$

$$\Rightarrow \boxed{a = 5}$$

$$\text{So } b^2 = 4 \times 5$$

$$\Rightarrow b = 2\sqrt{5}$$

Therefore the required hyperbola is:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{25} - \frac{y^2}{20} = 1$$

$$\Rightarrow \boxed{4x^2 - 5y^2 = 100}$$

Hence the result.

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(\pm 4, 0)$, the latus rectum is of length 12.

h/w

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Vertices $(\pm 4, 0)$, $e = \frac{4}{3}$.

$$a = 4$$

$$e = \frac{4}{3}$$

$$e^2 = \frac{16}{9}$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{a^2 + b^2}{a^2} = \frac{16}{9}$$

$$\frac{16 + b^2}{16} = \frac{16}{9}$$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(0, \pm \sqrt{10})$, passing through $(2, 3)$

(Sol) let the given hyperbola be:

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 \quad \text{--- (1)}$$

Since $(2, 3)$ lies on (1), so, we get

$$\frac{9}{a^2} - \frac{4}{b^2} = 1$$

$$\Rightarrow 9b^2 - 4a^2 = a^2b^2 \quad \text{--- (2)}$$

Also, It's given that $c = \sqrt{10}$

$$\Rightarrow c^2 = 10$$

$$\Rightarrow a^2 + b^2 = 10$$

$$\Rightarrow b^2 = 10 - a^2 \quad \text{--- (2)}$$

Using (2) in (1), we get;

$$9(10 - a^2) - 4a^2 = a^2(10 - a^2)$$

$$\Rightarrow 90 - 9a^2 - 4a^2 = 10a^2 - a^4$$

$$\Rightarrow 90 - 13a^2 = 10a^2 - a^4$$

$$\Rightarrow a^4 - 23a^2 + 90 = 0$$

Put $a^2 = t$, we get;

$$t^2 - 23t + 90 = 0$$

$$\Rightarrow t^2 - 18t - 5t + 90 = 0$$

$$\Rightarrow t(t - 18) - 5(t - 18) = 0$$

$$\Rightarrow (t - 18)(t - 5) = 0$$

$$\Rightarrow t = 18, 5$$

$$\Rightarrow a^2 = 18, 5$$

We reject $a^2 = 18$ as it cannot be possible because of $\neg$ (2).

$$\text{So } a^2 = 5$$

$$\Rightarrow b^2 = 10 - 5 \text{ (from ②)}$$

$$\Rightarrow b^2 = 5.$$

Therefore from ①, we get;

$$\frac{y^2}{5} - \frac{x^2}{5} = 1$$

$$\Rightarrow \boxed{y^2 - x^2 = 5} \text{ hence the result.}$$

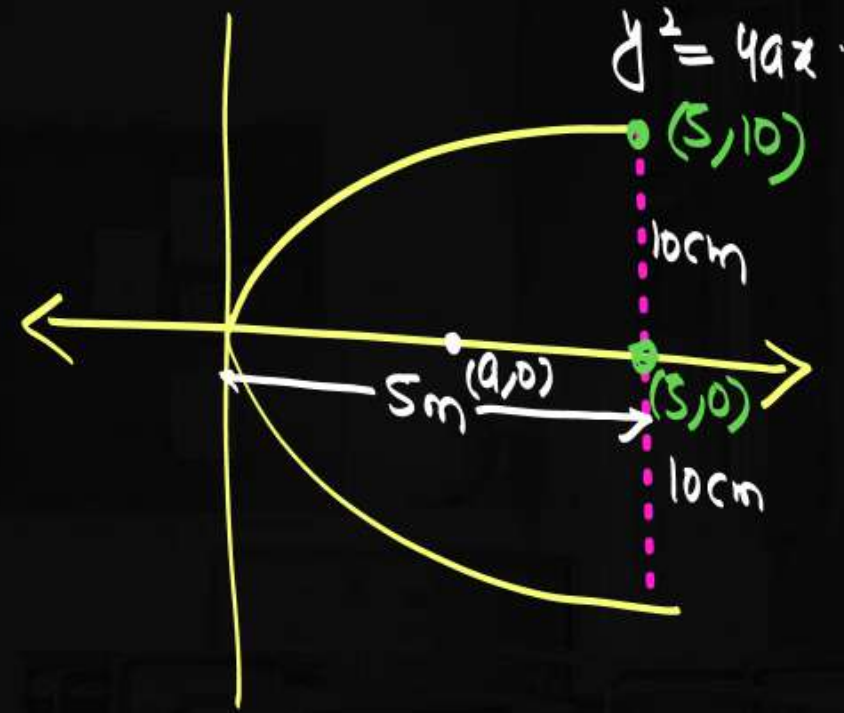
Homework

**MISCELLANEOUS
EXERCISE ON
CHAPTER – 10**

QUESTION

If a parabolic reflector is 20 cm in diameter and 5 cm deep, find the focus.

$$\begin{aligned}
 y^2 &= 4ax \\
 (10)^2 &= 4a \times 5 \\
 \Rightarrow 100 &= 20a \\
 \Rightarrow a &= 5.
 \end{aligned}$$



QUESTION

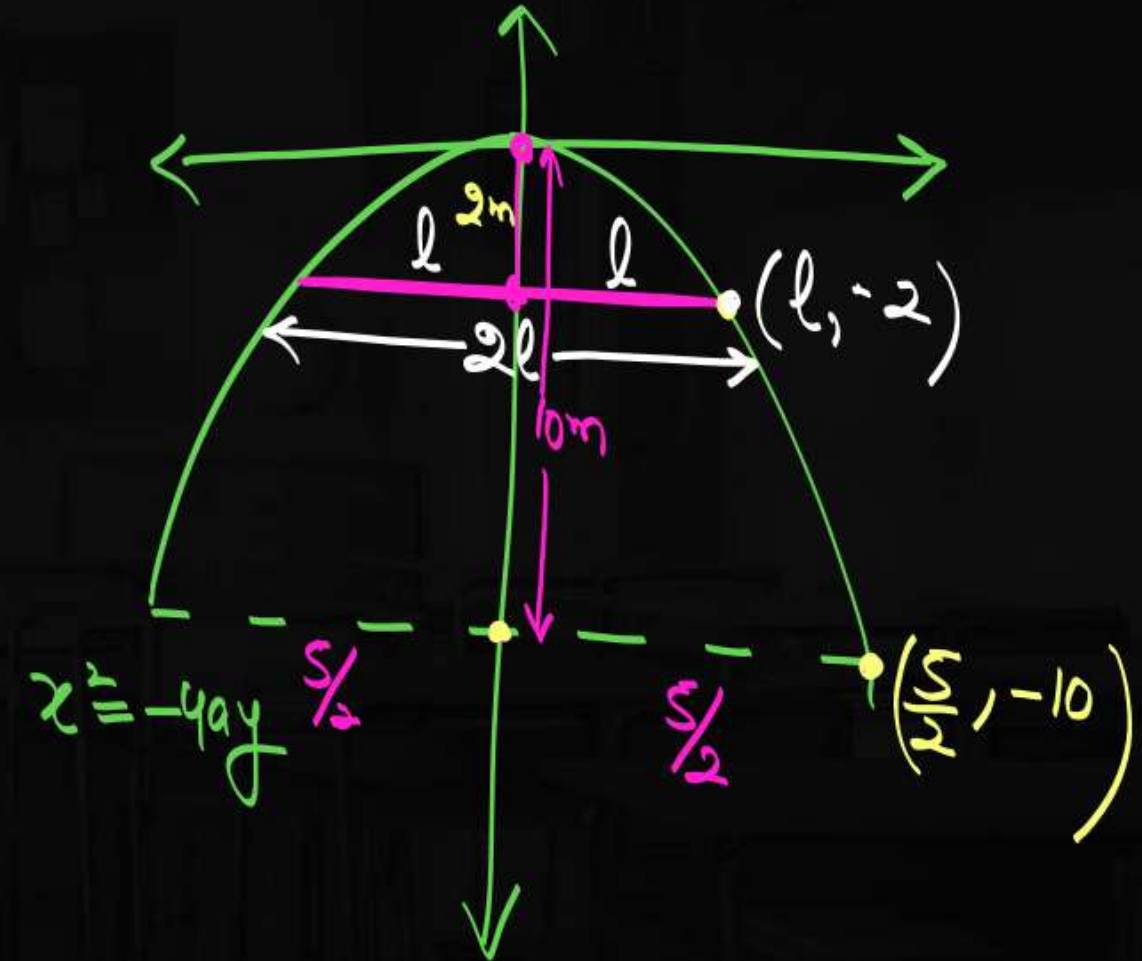
An arch is in the form of a parabola with its axis vertical. The arch is 10 m high and 5 m wide at the base. How wide is it 2 m from the vertex of the parabola?

(Sol)

$$\begin{aligned}
 x^2 &= -4ay \\
 \left(\frac{5}{2}\right)^2 &= -4ax - 10 \\
 &= 40a \\
 \frac{25}{4} &= 40a \\
 \boxed{a = \frac{5}{32}}
 \end{aligned}$$

$$\begin{aligned}
 x^2 &= -4\left(\frac{5}{32}\right)y \\
 \Rightarrow 32x^2 + 20y &= 0.
 \end{aligned}$$

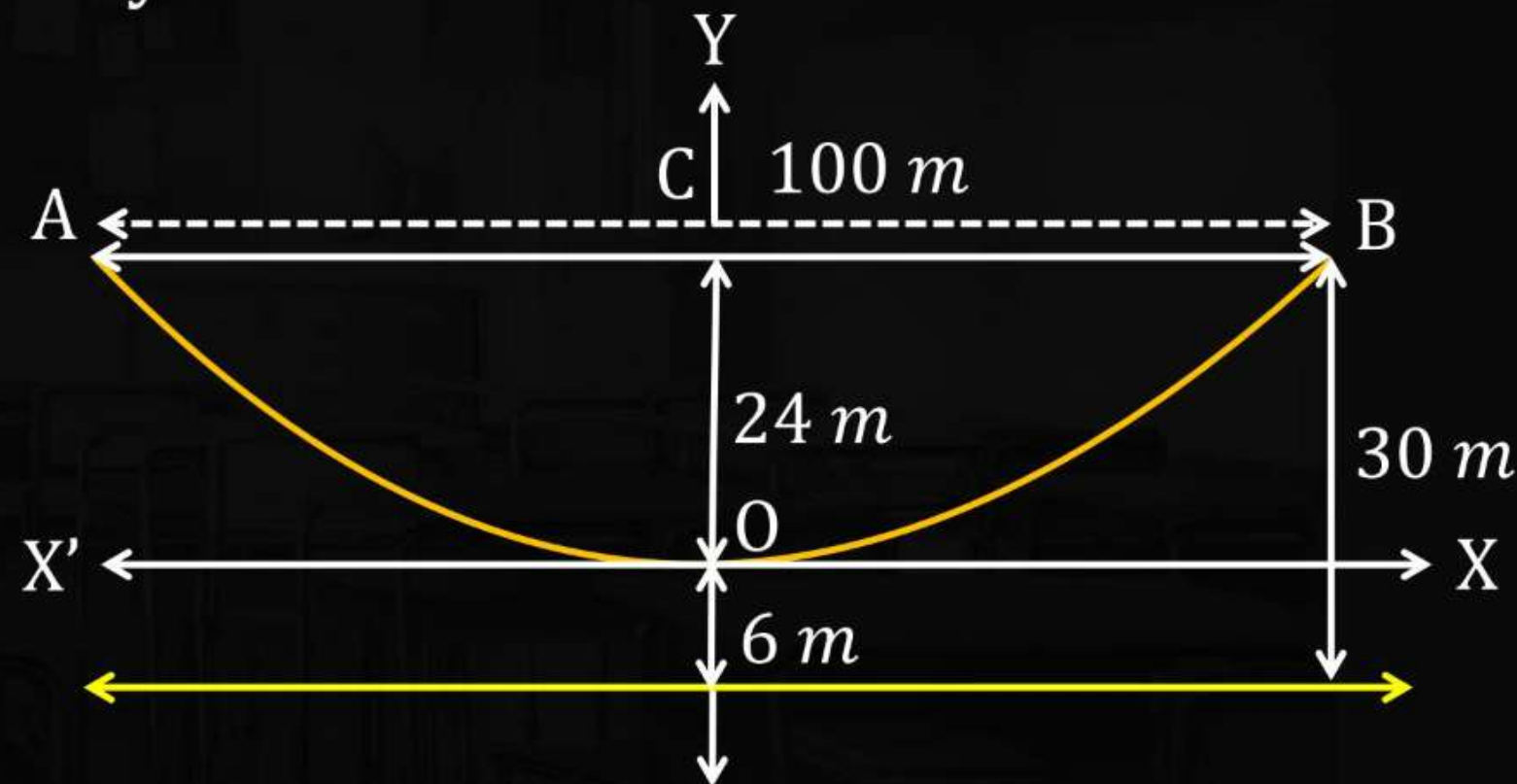
H/W



QUESTION

The cable of a uniformly loaded suspension bridge hangs in the form of a parabola. The roadway which is horizontal and 100 m long is supported by vertical wires attached to the cable, the longest wire being 30 m and the shortest being 6 m . Find the length of a supporting wire attached to the roadway 18 m from the middle.

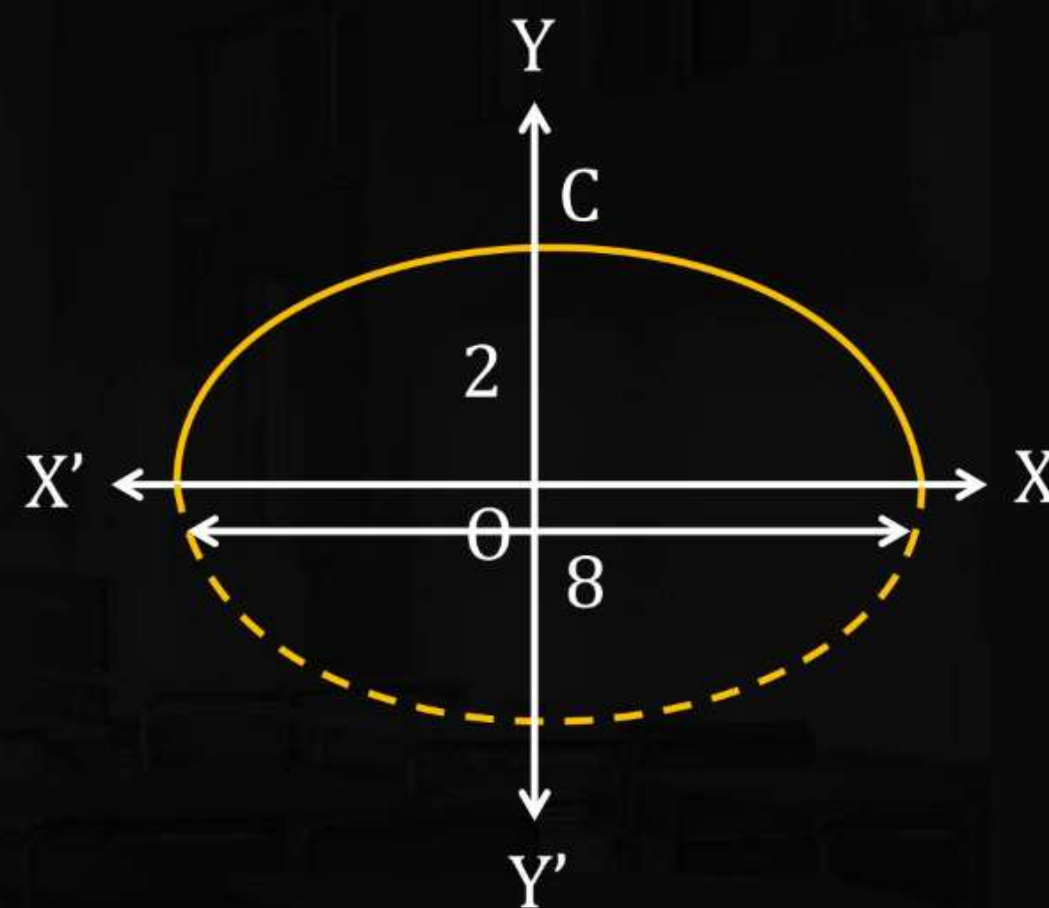
HP/W



QUESTION

An arch is in the form of a semi-ellipse. It is 8 m wide and 2 m high at the centre. Find the height of the arch at a point 1.5 m from one end.

h/w



QUESTION

A rod of length 12 cm moves with its ends always touching the coordinate axes. Determine the equation of the locus of a point P on the rod, which is 3 cm from the end in contact with the x -axis.

H/W

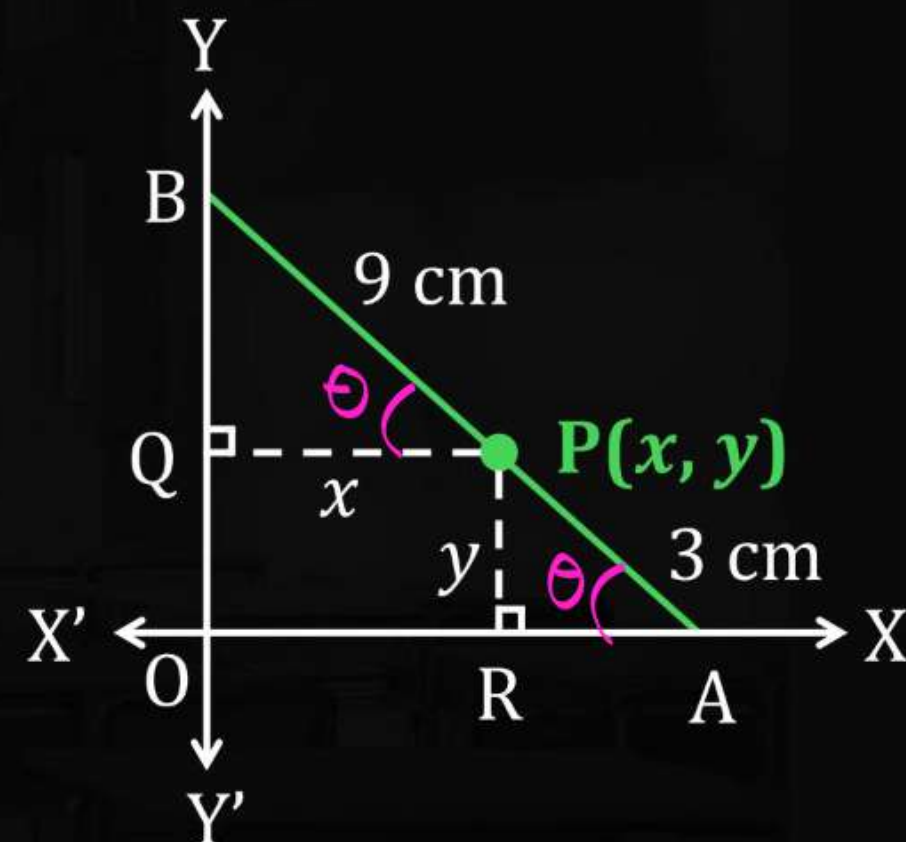
$$\frac{x^2}{9^2} + \frac{y^2}{3^2} = 1$$

$$y = 3 \sin \theta$$

$$x = 9 \cos \theta$$

$$\frac{y}{3} = \sin \theta$$

$$\frac{x}{9} = \cos \theta$$



QUESTION

Find the area of the triangle formed by the lines joining the vertex of the parabola $x^2 = 12y$ to the ends of its latus rectum.

Homework.

QUESTION

A man running a racecourse notes that the sum of the distances from the two flag posts from him is always 10 m and the distance between the flag posts is 8 m . Find the equation of the posts traced by the man.

QUESTION

An equilateral triangle is inscribed in the parabola $y^2 = 4ax$, where one vertex is at the vertex of the parabola. Find the length of the side of the triangle.

$$OC^2 + AC^2 = OA^2$$

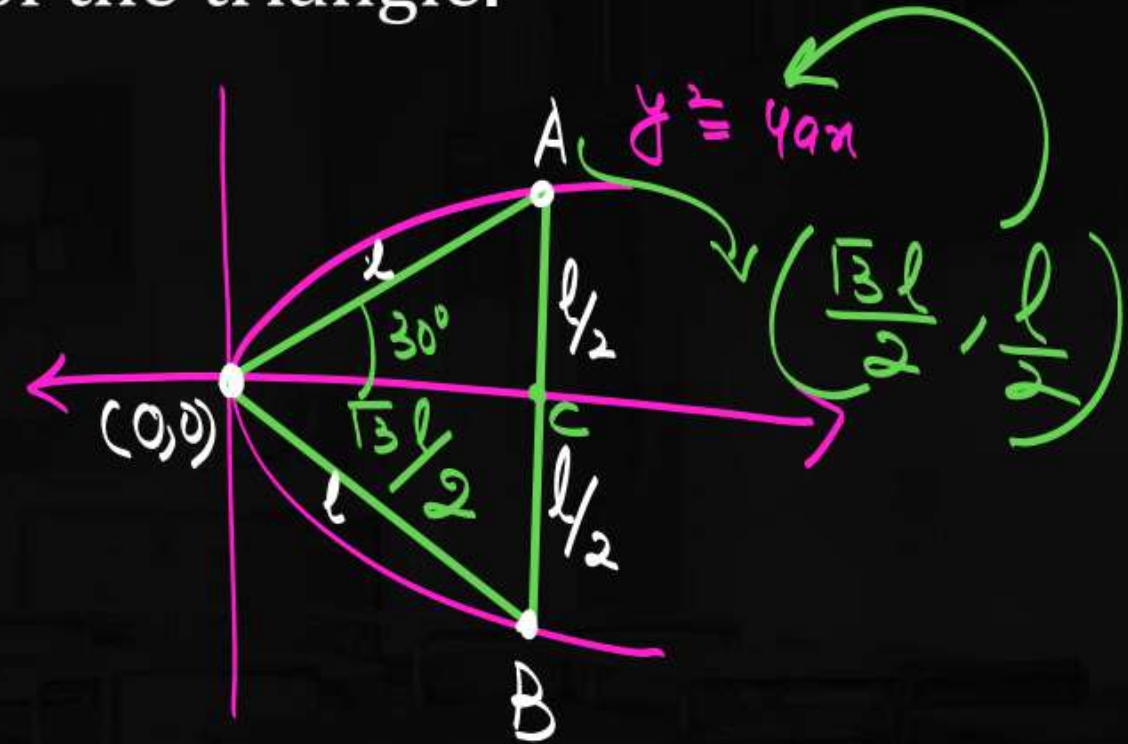
$$OC^2 = l^2 - \frac{l^2}{4} = \frac{3l^2}{4}$$

$$OC = \frac{\sqrt{3}l}{2}$$

$$\frac{l^2}{4} = 4a \times \frac{\sqrt{3}l}{2}$$

$$\frac{l}{4} = 2\sqrt{3}a$$

$$\Rightarrow l = 8\sqrt{3}a$$



Thank
You